



Hybrid multigroup partial least squares structural equation modelling: an application to bank employee satisfaction and loyalty

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Abstract

We describe a practical approach to tackling observed heterogeneity using partial least squares structural equation modelling (PLS-SEM) when the number of categorical variables is high and the context of the research is exploratory. The approach is based on combining classical multigroup PLS-SEM approach and pathmox analysis. We provide practical guidance on using our hybrid multigroup PLS-SEM and illustrate its application using real data for bank employees. In investigating work climate, specifically the relationship between satisfaction and loyalty considering specific drivers (empowerment, company reputation, leadership, pay, and work conditions) and different sources of heterogeneity (gender, age, marital status, education, job level, and antiquity), the hybrid multigroup PLS-SEM identified three partitions defined by juniors, seniors, and managers, and identified significant differences between those groups, specifically in indicating that leadership and pay were more important for juniors, empowerment for seniors, and company reputation and work conditions for managers.

Keywords Heterogeneity · Multigroup analysis · PLS-MGA · Pathmox analysis · Job satisfaction · Employee loyalty

1 Introduction

In recent decades, partial least squares structural equation modelling (PLS-SEM) has become well established as a multivariate method for analysing complex causal relationships between variables (Akter et al. 2017; Hair et al. 2018; Hair et al. 2014b; Sarstedt et al. 2014).

PLS-SEM allows multiple blocks of observed variables (or indicators) to be analysed, with each block playing the role of a latent variable when it is hypothesized that a system of linear relationships exists between blocks (Esposito Vinzi et al. 2010). Two

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models are computed: the measurement (or outer) model that relates observed variables to latent variables, and the structural (or inner) model that computes the strength and direction of relationships among the latent variables (Ciavolino et al. 2019). This approach is different from common factor-based (i.e., covariance-based) structural equation modelling (SEM), which, considering latent variables to be common factors that explain co-variation between the associated indicators (Sarstedt et al. 2016), is based on strict assumptions regarding distributions and sample size. PLS-SEM, in contrast, maximizes the variation in the observed variables following a soft approach that requires no strong assumptions regarding distributions, sample size (Hair et al. 2018).

PLS-SEM usefulness for prediction and the fact that no assumptions are required (Chin et al. 2020; Hair et al. 2012; Hwang et al. 2020; Reinartz et al. 2009) have contributed to its popularity and to its application in fields as varied as marketing (Cheah et al. 2020; Cheung et al. 2020; Lim et al. 2019; Osakwe et al. 2020), management (Chin et al. 2020; Hair et al. 2014a, b; Menon et al. 2020), sport (Chih-Hung Wang and Tang 2018; Howat and Assaker 2013), technology (Kim et al. 2020, Hensler et al. 2016a; Shiau et al. 2019; Lamberti et al. 2020a) and, most recently, psychology (Signore et al. 2019; Macchitella et al. 2020).

The problem remains, however, as to how to apply PLS-SEM in a heterogeneity context. While the emphasis may be on collecting data on the variables corresponding directly to the questions to be answered by a study, it is also usual to collect background information, e.g., on sociodemographic variables (gender, age, social status, education, income, etc.) that may be useful in segmenting groups and identifying different patterns in models, i.e., heterogeneity. Using a single model when heterogeneity is present can be misleading as the causal relationships between constructs are likely to differ depending on differing group profiles defined by specific segmentation variables (Becker et al. 2013). Resolving the heterogeneity problem may require different analyses for heterogeneous and homogeneous groups, as defined by segmentation variables.

Researchers have invested considerable efforts in developing methods capable of taking observed and unobserved (or latent) heterogeneity into account in stepped model estimates (Becker et al. 2013; Chin 2000, 2003; Esposito Vinzi et al. 2008; Fordellone and Vichi 2020; Henseler et al. 2009; Keil et al. 2000; Lamberti et al. 2016; Ringle et al. 2010; Ringle and Schlittgen 2007; Schlittgen et al. 2016; Tenenhaus et al. 2006). Focusing (for our purposes) on observed heterogeneity, a widely used approach is multigroup analysis (MGA), with applications reported for a multitude of contexts, ranging from marketing to the social sciences (Hair 2018; Matthews 2017; Sarstedt et al. 2011). The principal reasons for its popularity are that it is simple to use and easy to interpret, and has suitable statistical modelling software available, including R (R Development Core Team 2012), XLSTAT (Esposito Vinzi et al. 2007) and SmartPLS (Sarstedt and Cheah 2019; Ringle et al. 2005). However, MGA is less than efficient in three scenarios: (1) when there is a large number of segmentation variables, (2) when there is a large number of segmentation levels, and—related to these issues—(3) when the research aim is to consider the mixed effects of segmentation variables. Those three factors are especially relevant when we are analysing survey-based data with numerous sociodemographic details for respondents.

Pathmox analysis (Lamberti et al. 2016, 2017) was developed as a possible solution to this problem. Using an iterative procedure, all possible binary partitions of the segmentation variables are analysed by an algorithm and significant segments are generated as output. However, the method has the drawback that it cannot verify, *a posteriori*, whether the path coefficients of the defined segments are statistically significant.

As a solution to overcome the limitations of the MGA and pathmox approaches to observed heterogeneity when the number of segmentation variables is high, we propose a method that we call hybrid multigroup PLS-SEM in our article, organized as follows. We first describe the main approaches to dealing with observed heterogeneity and provide practical guidance on applying MGA and pathmox analysis. We then describe our hybrid multigroup approach and illustrate it with an application to satisfaction and loyalty among the employees of a large Spanish bank. Finally, we conclude with a discussion of our findings and our conclusions.

2 Heterogeneity in PLS-SEM

Classically, depending on a priori knowledge we can distinguish between two kinds of heterogeneity: (1) observed heterogeneity, where the number of variables is known a priori (e.g., if we know that heterogeneity is defined by age and education we can define segments accordingly); and (2) unobserved (or latent) heterogeneity, where the number of segments and their composition is not known a priori but becomes apparent from differences in structural path coefficients. Here we focus on observed heterogeneity for the MGA and pathmox analysis, underlining the corresponding advantages and limitations.

2.1 Multigroup analysis

MGA, which was developed for PLS-SEM, tests statistical differences in path coefficients between segments. Heterogeneity is usually observed, i.e., the number of groups in which observations can be segmented is known a priori; indeed, most of the MGA applications in the literature are based on a limited nominal number of segmentation variables observed as sources of heterogeneity. The data is separated into segments according to segmentation variables and calculated for each segment are path coefficients, which are then compared to identify significant differences. This comparative analysis is based on testing two hypotheses: the null hypothesis, which is that path coefficients between segment are the same, i.e., $\beta_{G1}^{path} = \beta_{G2}^{path}$, and the alternative hypothesis, which is that path coefficients between segments are different, i.e., $\beta_{G1}^{path} \neq \beta_{G2}^{path}$. The statistics used to test the null hypothesis can be calculated using several procedures.

Tests reported in the literature and exhaustively reviewed by Hair et al. (2018), include the parametric t-test (Keil et al. 2000; Chin 2000) along with its Smith-Satterthwaite adaptation, the PLS-MGA (Henseler et al. 2009), and the permutation test (Chin 2003). An additional method is the omnibus test proposed by Sarstedt et al. (2011), which allows the comparison to be extended to more than two groups. The parametric t-test and Satterthwaite adaptation both use a bootstrap re-sampling procedure; the coefficients for each segment are calculated in each re-sampling and the standard error estimates are parametrically tested using t-tests. The PLS-MGA applies a non-parametric approach, and, like the parametric approach, uses a bootstrap procedure. For each parameter, bootstrap estimates for each segment are compared with bootstrap estimates for other segments, and counting the number of times the bootstrap estimate for a first segment is larger than that of a second segment obtains a p-value for a one-tailed test. The permutation test, which applies approximate randomization (i.e., permutation) to evaluate path coefficient differences across segments, operates on the basic premise that all segments are equivalent, and that every segment member is the same as before sampling began. A calculated statistic is determined to

be significant on the basis of the likelihood of its significance if assignments to segments have been jumbled/mixed. Finally, the omnibus test combines bootstrapping and permutation testing to mimic an overall F-test.

The main limitation of these approaches is in relation to dimensionality: making comparisons and understanding the reasons for significant differences become increasingly difficult as the number of segments increases. This happens whenever there is a large number of segmentation variables, when variables have a large number of levels and/or when effects are mixed and models are complex.

Regarding software and the tests they implement, SmartPLS implements the parametric t-test (Chin 2000; Keil et al. 2000), the Smith-Satterthwaite t-test adaptation and the PLS-MGA (Henseler et al. 2009), whereas the *plspm* package in R (Sanchez et al. 2013) and XLSTAT implement the parametric t-test (Chin 2000; Keil et al. 2000) and the permutation test (Chin 2003).

2.2 Pathmox analysis

Pathmox analysis (Lamberti et al. 2016, 2017) has recently been proposed to analyse heterogeneity in the PLS-SEM environment, as it is capable of detecting heterogeneity when many segmentation variables are present and when it is difficult to determine which variables are responsible for differences. The principles of binary segmentation are applied in pathmox analysis to produce a tree with different models in each of the obtained nodes. Pathmox functions as follows:

- Step 1 A global PLS-SEM model is fit to all the data to define the root of the tree.
- Step 2 The pathmox algorithm establishes the set of admissible partitions for each node in the tree considering the available segmentation variables.
- Step 3 In an iterative procedure, models with the greatest differences in each child node are identified.

Three parameters need to be pre-defined for the algorithm: (1) the significance threshold, which, according to classical statistical testing, can be fixed as $p=0.05$; (2) the minimum number of elements per node, which depends on sample size, which, in turn, should be large enough to guarantee the goodness of the results; and (3) the depth of the tree's growing level, given that the maximum tree level has to be moderate to avoid overfitting, and this also ensures that the algorithm does not have to search for large trees with large numbers of leaves and that the final number of interpretable segments is manageable.

Optimal partitions and binary splits are determined by a statistical test based on Fisher's F-test for equality in regression models (Chow 1960; Lebart et al. 1979), adapted for two structural models calibrated from different segments (successors of a node). This test, in model comparisons at the overall structural level, compares parent node path coefficients with child node path coefficients as defined by the binary splits. The parent node model reflects homogeneity, whereas the child node models reflect heterogeneity.

The main limitation of pathmox analysis is related to its goal, which is to identify the most significant models differing in terms of available sources of heterogeneity. However, no a posteriori test can verify whether significant differences, if any, exist between the path coefficients for the identified models.

Pathmox analysis can be implemented using the `genpathmox` package in R (Lamberti 2014) using the `pls.pathmox` function. A technical sheet is available on the website describing the necessary steps and R code for definition of the hybrid segmentation variable.

3 Hybrid multigroup PLS-SEM

To overcome the limitations of MGA and pathmox analysis we propose our hybrid multigroup approach, which involves three steps that can be summarized as follow:

Step 1

Determine the hybrid segmentation variable as follows

- a. Perform pathmox analysis to identify the most significant subgroups
- b. Define the hybrid segmentation variable according to the resulting final tree nodes
- c. Interpret the levels of the hybrid segmentation variable according to the segmentation variables identified as responsible for the tree partitions.

Step 2 Estimate and analyse the PLS-SEM models associated with the hybrid segmentation variable levels

Step 3 Run MGA comparisons of path coefficients for each group using one of the following tests: parametric test, permutation test, PLS-MGA test or omnibus test

The hybrid segmentation variable is first built as a categorical variable where levels identify more different PLS-SEM models associated with more different subgroups of individuals. Next, using pathmox analysis, subgroups are identified by partitioning data according to the available segmentation variables. The resulting subgroups are then compared to check if they are significantly different and, if so, the most significant subgroups are identified. Each identified subgroup is therefore associated with one level of the hybrid segmentation variable. Ultimately, the final tree nodes identified by pathmox analysis define the hybrid segmentation variable, while each node identifies a different level of the hybrid segmentation variable. The next stage is to determine the meaning of the levels of the hybrid segmentation variable, which requires a suitable label to be assigned to each variable level. This is done by analysing the segmentation variables that the pathmox analysis identified as responsible for the tree partitions and assigning a suitable label to the level associated with each terminal node. Once the hybrid segmentation variable has been interpreted, the PLS-SEM models associated with each variable level can be estimated and MGA can be implemented in order to identify the statistically different coefficients, using an MGA test selected in accordance with the literature (see Hair et al. 2018).

Before implementing the MGA, it is recommended to test measurement invariance, using, for instance, the measurement invariance of composite models (MICOM) procedure (Henseler et al. 2016a, b).

Measurement invariance is a statistical property of measurement that demonstrates whether or not we are measuring the same constructs in various specified groups. As argued by Hair et al. (2018) and Henseler et al. (2016a, b), the establishment of measurement invariance across groups is a logical prerequisite to conducting group comparisons and necessary to ensure that model estimate differences do not result from distinctive content or meanings for the unobserved variables across groups (Hair et al. 2018).

4 Application to bank employee data

Using work climate data for the financial sector, we illustrate how hybrid multigroup PLS-SEM deals with heterogeneity when there is a large number of segmentation variables. The five segmentation variables used in this case were age, gender, marital status, education, job level (position in the company hierarchy), and antiquity (length of service). The data and model have recently been published (see Lamberti et al. 2020b).

In the financial sector, the impact of the work climate on the relationship between strategic human resource (HR) management and organizational performance is especially crucial (Abdolshah et al. 2018; Bartel 2004; Clark 2002; Kaya et al. 2010; Richard and Johnson 2001; Sekaran 1989; Tlaiss 2013; among others). Another issue of relevance is that different groups of employees may respond in different ways to specific HR management practices (Eskildsen et al. 2004; Tlaiss 2013; Valaei and Jiroudi 2016).

Our objectives were (1) to identify key characteristics defining different groups of employees, and (2) to analyse differences in the models of those groups, i.e., differences in the causal relationships between the latent variables. We started by considering a holistic model that related employee satisfaction and loyalty with the five work climate constructs of empowerment, company reputation, pay, work conditions, and leadership (Eskildsen et al. 2004, 2010a, b; Gaertner 1999; Kristensen and Westlund 2004; Lane et al. 2010). We then tested all the relationships against the six sociodemographic variables of age, gender, marital status, education, job level, and antiquity.

In this case the number of variables is high, so use of classical MGA would be complex and produce inaccurate results, whereas use of pathmox analysis could result in significant groups but would not identify the significant differences between them.

4.1 Data

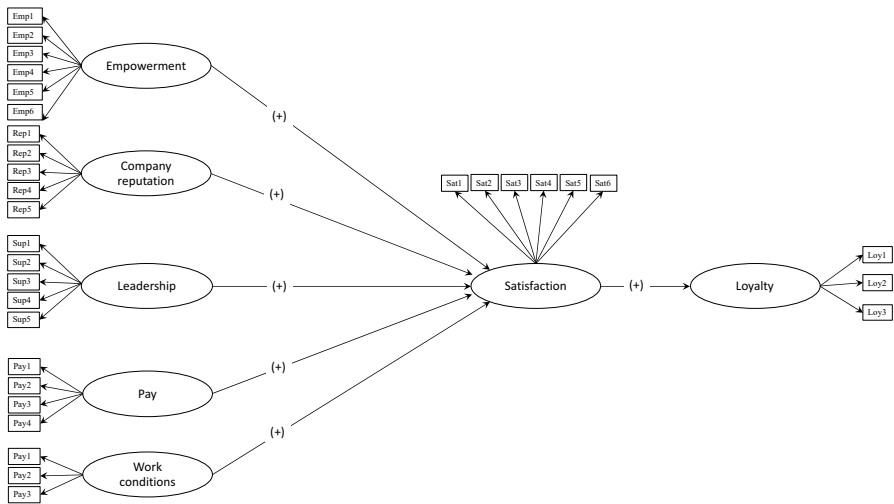
Our data,¹ obtained from an organizational study of an international bank, consisted of 37 variables for 2000 employees divided into six qualitative variables and 31 indicator (or manifest) variables. The six qualitative variables, reflecting specific employee characteristics that served as potential observed sources of heterogeneity (segmentation variables) were age, gender, marital status, education, job level, and antiquity. The 31 indicator (or manifest) variables were satisfaction (five indicators), loyalty (three indicators), and the five work climate constructs of empowerment (six indicators), company reputation (five indicators), leadership (five indicators), pay (four indicators), and work conditions (three indicators) (see description and summary statistics in Appendix). Each indicator was measured on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree).

The employees in the sample were mostly men (69.70%) and were mostly younger than 45 years old (80.55%). Around 50% were married, most had graduate studies (59.65%), and a third (30.20%) had post-graduate studies. Low-level employees followed by high-level employees represented around two third each of the sample (39.45% and 35.10%, respectively), and most employees (72.45%) had been working in the bank for more than five years. Summarized descriptive information for the employees is provided in Table 1.

¹ The data were obtained from an organizational study of a Spanish bank collaborating with a Spanish university statistics department. The data had been collected through a survey (run by the bank's human resource department), consisting of a questionnaire emailed to employees, whose anonymity was guaranteed. The response rate was 94%. The collected data were processed and analyzed by the university statistics department.

Table 1 Descriptive details of 2,000 bank employees

Gender	%	Age	%	Job level	%
Female	30.30	< 31 years	33.45	Low	39.45
Male	69.70	31–45 years	47.10	Intermediate	25.45
		> 45 years	19.45	High	35.10
Antiquity	%	Education	%	Marital status	%
< 2000	10.30	Secondary	9.15	Not married	49.50
2000–2004	23.50	Graduate	59.65	Married	50.50
2005–2009	19.75	Post-graduate	31.20		
2009–2014	18.90				
> 2014	27.55				

**Fig. 1** Work climate model

4.2 Model

Our model, depicted in Fig. 1, is similar to/an adaptation of models described elsewhere (Eskildsen et al. 2004, 2010a, b; Gaertner 1999; Lane et al. 2010). The model reflects employee satisfaction and loyalty (the endogenous constructs) and their antecedents, namely, empowerment, company reputation, pay, work conditions, and leadership (the exogenous constructs), i.e., latent variables assumed to affect the work climate. Table 2 summarizes the definitions of each construct, their hypothesized effect on other constructs, and the corresponding literature references, while a full explanation of the causal relationships can be found in Lamberti et al. (2020b).

Table 2 Constructs, hypothesized effects and literature references

Construct	Definition	Hypothesis	References
Empowerment	Employee perceptions of management vision (vertical or horizontal)	Positive effect on satisfaction	Mahmood et al. (2019), Ullah and Yasmin (2013)
Reputation	Employee perceptions of the company's reputation	Positive effect on satisfaction	Herrbach et al. (2004), Helm (2011)
Leadership	Employee perceptions of leadership practices such as feedback and appraisal	Positive effect on satisfaction	Ugboro and Obeng (2000), Puni et al. (2018)
Pay	Employee perceptions of pay and of its equity	Positive effect on satisfaction	Guan et al. (2014); Mahmood et al. (2019)
Conditions	Employee perceptions regarding the conditions in which they perform their work	Positive effect on satisfaction	Giaveli and Karassavidou (2011), Kaya et al. (2010)
Satisfaction	Employee contentedness with their job and aspects such as kind of work, supervision, etc	Positive effect on loyalty	Eskildsen et al. (2004), Dhir et al. (2020)
Loyalty	Employee's level of attachment or commitment to their company		Guillon and Cezanne (2014)

4.3 Softwares

The SmartPLS 3.2 software (Sarstedt and Cheah 2019) and the *genpathmox* package in R software version 0.4 (Lamberti 2014) provided the model estimates (global and multigroup analysis) and the pathmox analysis, respectively.

4.4 Results

4.4.1 Global PLS-SEM

All the latent variables in the model were modelled assuming that each was an antecedent of its own indicators (i.e., a reflective approach). It means that it was assumed that manifest variables were considered to be caused by latent variables. This approach differs from the formative approach, in which the latent construct is assumed to be formed by its indicators (Cheah et al. 2020).

The model was validated, following indications by Ringle et al. (2018) for PLS-SEM model use in the HR management field, and common reliability indexes were computed for each latent variable. The results are presented in Table 3. In line with Henseler et al. (2016a, b), the reliability of a group was considered to be satisfactory if composite reliability (CR), Cronbach's α and ρ_A were greater than 0.7. CR values were between 0.808 and 0.932 for all the constructs. Cronbach's α values were higher than 0.70, except for work conditions (0.692) and loyalty (0.659), while ρ_A (Dijkstra and Henseler 2015) values were all higher than 0.70, except (again) for work conditions (0.695) and loyalty (0.698). We retained those constructs, however, as the values were close to the 0.7 threshold and loadings were higher than 0.7. Internal consistency for all the groups was supported by average variance extracted (AVE). All item loadings were close to or higher than the 0.7 threshold except few cases. Again, we maintained these as reliability and internal consistency was good enough and it seemed important to avoid reducing richness in the construct indicators. All loadings were significant according to the t-test. To assess discriminant validity, we calculated the heterotrait-monotrait (HTMT) ratio of correlations (Henseler et al. 2015), which can be interpreted as the correlation between two constructs; its maximum value, which should be as low as possible, was 0.839 in our case, i.e., lower than the conservative threshold of 0.850.

Common method bias (CMB) was tested following the approach described by Valaei and Jiroudi (2016) for a job satisfaction model applied to a HR management context and validated using Harman's one-factor test. We entered all principal constructs into a single principal component factor analysis, setting to one the number of factors. Results show that only one factor emerged, explaining 40% of the variance, which, at under 50%, indicated the absence of CMB. We also ran the full collinearity test (Kock 2015), consisting of calculating all variance inflation factors (VIFs) for the latent variables and confirming that they were lower than 3.3. Table 4 report the full collinearity test results.

The structural model results are reported in Table 5 and Fig. 2. The main drivers for satisfaction were empowerment ($\beta=0.540$), followed by company reputation ($\beta=0.168$), work conditions ($\beta=0.130$), and pay ($\beta=0.118$), with the value for leadership ($\beta=0.044$) indicating it to be the least important driver. All coefficients were significant according to the t-test. Concerning the predictability of the model, in line with Ringle et al. (2018), we calculated R^2 and the Stone-Geisser test of predictive relevance (Q^2), obtaining R^2 values of 0.591 and 0.437 and Q^2 values of 0.537 and 0.341 for satisfaction and loyalty,

Table 3 Outer model results for 31 indicators corresponding to five constructs

Constructs and indicators	Loadings	t*	Cronbach's α	ρ_A	CR	AVE				
<i>Empowerment</i>										
Empo1	0.793	114.175	0.924	0.841	0.515					
Empo2	0.799	128.854								
Empo3	0.752	92.468								
Empo4	0.612	41.841								
Empo5	0.657	50.403								
Empo6	0.636	44.846								
<i>Reputation</i>										
Rep1	0.751	49.700	0.909	0.914	0.932	0.733				
Rep2	0.665	38.920								
Rep3	0.739	47.317								
Rep4	0.690	37.068								
Rep5	0.739	47.800								
<i>Leadership</i>										
Lead1	0.849	156.140	0.766	0.770	0.851	0.590				
Lead2	0.826	113.878								
Lead3	0.897	238.507								
Lead4	0.850	156.956								
Lead5	0.858	158.617								
<i>Pay</i>										
Pay1	0.818	103.657	0.695	0.830	0.620					
Pay2	0.659	44.722								
Pay3	0.749	60.790								
Pay4	0.835	106.675								
<i>Conditions</i>										
Work1	0.798	91.887					0.808	0.808	0.867	0.566
Work2	0.828	112.363								
Work3	0.734	59.499								
<i>Satisfaction</i>										
Sat1	0.725	76.799	0.659	0.698	0.808	0.584				
Sat2	0.781	98.479								
Sat3	0.736	77.744								
Sat4	0.781	101.329								
Sat5	0.737	72.471								
<i>Loyalty</i>										
Loy1	0.825	129.954	0.659	0.698	0.808	0.584				
Loy2	0.723	58.706								
Loy3	0.742	61.590								

*t-test significance values

Table 4 Full collinearity test results for common method bias

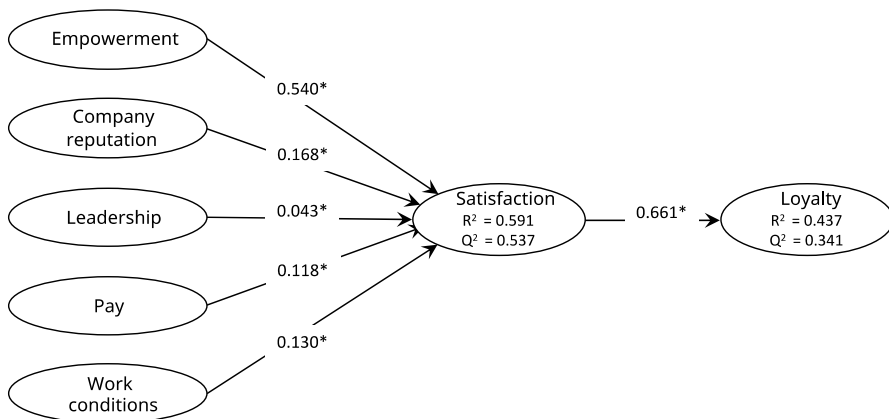
Construct	Empowerment	Image	Pay	Work condition	Leadership	Satisfaction	Loyalty
Empowerment		1.445	1.218	1.363	1.244	2.220	2.062
Image	2.667		1.205	1.393	1.460	2.698	1.901
Pay	2.666	1.430		1.368	1.460	2.665	2.134
Work condition	2.606	1.442	1.194		1.460	2.671	2.125
Leadership	2.272	1.444	1.217	1.395		2.716	2.135
Satisfaction	2.172	1.430	1.190	1.367	1.455		1.920
Loyalty	2.574	1.285	1.216	1.388	1.459	2.451	

VIF < 3.3 indicates no common method bias

Table 5 Structural model results for specific path coefficients

Path coefficients	Path	t*	p value	Total effects
Empowerment → Satisfaction	0.54	38.943	< 0.001	0.357
Reputation → Satisfaction	0.168	15.326	< 0.001	0.111
Leadership → Satisfaction	0.044	3.351	< 0.001	0.029
Pay → Satisfaction	0.118	9.051	< 0.001	0.078
Conditions → Satisfaction	0.13	69.653	< 0.001	0.086
Satisfaction → Loyalty	0.661	9.592	< 0.001	
	R ²	Q ²		
Satisfaction	0.591	0.537		
Loyalty	0.437	0.341		

*t-test significance values



* p-value < 0.001

Fig. 2 Structural model results showing R² and Q² values

Table 6 PLSpredict results for satisfaction and loyalty

Constructs	PLS-RMSE	Q ² predict	LM-RMSE	Difference
Loy1	0.728	0.357	0.729	−0.001
Loy4	0.726	0.153	0.696	0.030
Loy5	0.738	0.201	0.739	−0.001
Sat1	0.559	0.360	0.560	−0.001
Sat2	0.658	0.347	0.663	−0.005
Sat3	0.609	0.372	0.610	−0.001
Sat4	0.689	0.326	0.678	0.011
Sat5	0.600	0.248	0.590	0.010

respectively. We also ran PLSpredict,² as indicated in Shmueli et al. (2016, 2019) (see Table 6) to compare PLS-SEM prediction for key dependent constructs and indicators with the prediction obtained using a naïve linear model (LM) approach that did not consider a causal relationship. According to Shmueli (2019), if the distribution of prediction errors can be assumed to be normal, then comparing the square root of the average of the squared differences between the predictions and the actual root mean square error (RMSE) observations between PLS-SEM and LM, we should find the RMSE for PLS-SEM to be lower than the RMSE for LM. In the best-case scenario, if the RMSE for PLS-SEM is lower in 100% of the indicators, then we can conclude that the model has high predictive power, while if no lower than 50% of the indicators, then we can assume that the model has moderate predictive power. As confirmed by the results in Table 6, our model has moderate predictive power.

Finally, concerning total effects on loyalty, the most important drivers were empowerment ($\beta = 0.351$), followed by company reputation ($\beta = 0.111$), work conditions ($\beta = 0.086$), pay ($\beta = 0.078$), and leadership ($\beta = 0.029$).

4.4.2 Hybrid multigroup PLS-SEM results

Step 1: Determining the hybrid segmentation variable

Pathmax analysis was implemented using the variables listed in Table 1. In terms of investigating possible differences in how job satisfaction, and thus loyalty, were perceived by the bank employees, the hybrid segmentation variable allowed us to identify the most important segments for comparison purposes. Given our objective of identifying only the main groups of employees, we limited the segmentation tree to a maximum depth of two levels and bounded the final number of segments to a maximum of four. The minimum admissible size for a node was set to 10% of the total sample, while the threshold significance for the partitioning algorithm was $p = 0.05$. The resulting pathmax tree, depicted in Fig. 3, had a total of three terminal nodes.

The algorithm selected job level as the variable with the greatest power ($p < 0.001$) to distinguish between higher- and lower-level employees. The lower-level employees were

² We fixed to one (i.e., $r = 1$) the number of repetitions. According to Shmueli et al. (2019), this is adequate when the prediction is based on a single model. We maintained to 10 the number subsample (i.e., $k = 10$) as this value respects the minimum required training sample size.

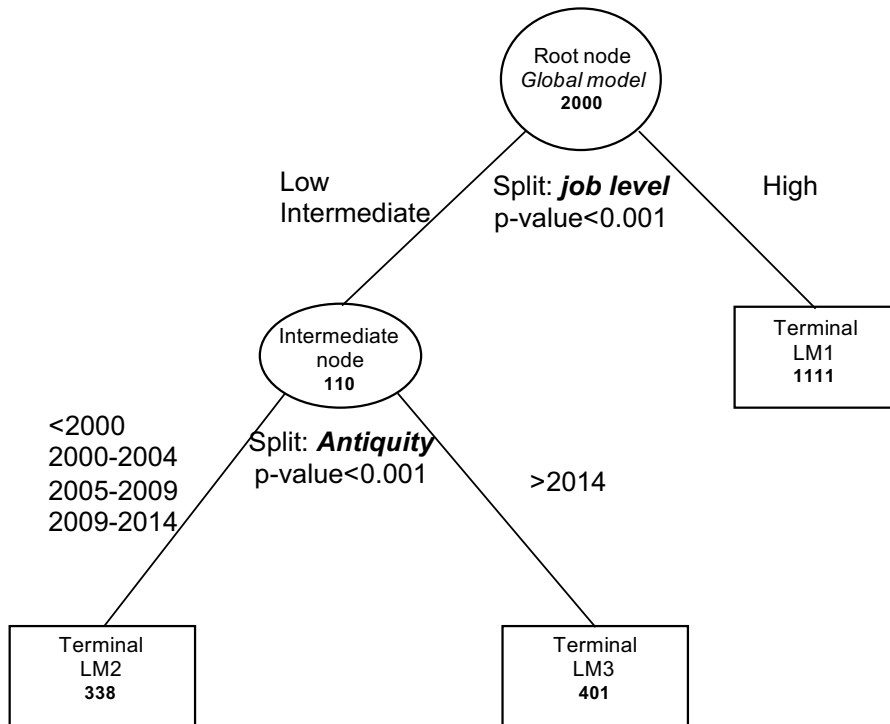


Fig. 3 Pathmox tree

further differentiated according to antiquity ($p < 0.001$). Identified, thus, were three segments as follows: higher-level employees (managers), lower-level employees with greater antiquity (seniors), and lower-level employees with lesser antiquity (juniors). These three segments corresponded to the three hybrid segmentation variable levels used subsequently in the MGA to compare path coefficients.

Step 2: Estimating and analysing the PLS-SEM models associated with the hybrid segmentation variable levels

Table 7 shows the path coefficients and the t-test statistics calculated for the groups according to the hybrid segmentation variable levels. The most important driver for satisfaction overall was empowerment, followed by leadership and pay for junior employees, and company reputation and work conditions for managers. As for seniors, relative to the other two groups, the most important driver for satisfaction was empowerment. Finally, satisfaction was important for loyalty in all three groups; nonetheless, although differences could be observed between the three groups, we needed to verify whether those differences were statistically significant, which was done using MGA.

Step 3: Multigroup comparisons of path coefficients

Measurement invariance test. Before making the group comparison, we checked measurement invariance following the three-step measurement invariance of composite models (MICOM) procedure (Henseler et al. 2016a, b), which ensures that model

Table 7 PLS-SEM model with hybrid segmentation variable levels

Path coefficients	Juniors			Seniors			Managers		
	Path	t*	p	Path	t*	p	Path	t*	p
Empowerment → Satisfaction	0.513	3.307	<0.001	0.65	4.666	<0.001	0.477	15.663	<0.001
Reputation → Satisfaction	0.088	4.484	<0.001	0.113	3.973	<0.001	0.21	10.385	<0.001
Leadership → Satisfaction	0.155	4.767	<0.001	0.025	0.871 ^{NS}	0.324	0.037	2.165	0.019
Pay → Satisfaction	0.143	4.79	<0.001	0.101	4.223	<0.001	0.072	4.159	<0.001
Conditions → Satisfaction	0.121	33.476	<0.001	0.103	33.628	<0.001	0.187	46.391	<0.001
Satisfaction → Loyalty	0.671	15.841	<0.001	0.658	22.568	<0.001	0.633	25.734	<0.001

*t-test significance values. NS indicates non-significant according to the t-test

estimate differences do not result from distinctive content or meanings for the latent variables across groups (Hair et al. 2018). The MICON procedure involves three steps: (1) configural invariance, (2) compositional invariance, and (3) equality of composite mean values and variance values. Following Hair et al. (2018), to apply the MGA, we needed to establish both configural invariance and compositional invariance so as to establish partial measurement invariance.

Configural invariance, which ensures that each latent variable has been specified the same for all the groups, exists when constructs are equally parameterized and estimated across groups. To establish configural invariance, each latent variable in the PLS-SEM model must therefore be specified equally for all the groups, ensuring, in our case, identical indicators for each measurement model, identical data treatment, and identical algorithm settings across groups. Compositional invariance means that composite scores measure exactly the same construct across groups.

The MICOM procedure tests for compositional invariance by calculating the latent score correlations between groups and comparing them with the reference distribution of correlations obtained by permutation of the two groups. If the observed correlation falls in the upper 95% of the distribution, then the null hypothesis that the theoretical correlation is 1 is accepted, and composite invariance of the specific construct is established. If, additionally, the constructs have equal mean values and variances across the groups, full measurement invariance is confirmed, allowing the data of different groups to be pooled.

Tables 8 and 9 show MICON step 2 and step 3 results for the observed correlations with the lowest 5% quantile of the reference distribution, the observed differences in means between the two compared groups, and the log ratio of variances for both groups with their corresponding 95% confidence intervals (obtained by group permutation). Compositional invariance was verified in most cases; in the very few exceptions, since the observed deviation was minimal we could globally assume the compositional invariance of the constructs. Regarding the equality of means of the composite scores, the fact that we could appreciate differences in most cases confirmed the existence of different segments according to our sources of heterogeneity. However, in terms of the variability of the composite scores for the segments, in many cases the variance between groups was similar. Since our results confirmed partial measurement invariance, we could proceed to MGA.

Table 8 MICON step 2 results

Construct	Group	Compositional invariance		
		Score correlation	5% quantile	Verified
Empowerment	Juniors versus managers	0.999	0.999	Yes
Reputation		0.988	0.988	Yes
Leadership		1.000	1.000	Yes
Pay		0.996	0.996	Yes
Conditions		0.999	0.997	Yes
Satisfaction		1.000	0.999	Yes
Loyalty		0.995	0.998	No
Empowerment	Juniors versus seniors	0.999	0.999	Yes
Reputation		0.993	0.983	Yes
Leadership		1.000	1.000	Yes
Pay		0.990	0.994	No
Conditions		1.000	0.997	Yes
Satisfaction		1.000	1.000	Yes
Loyalty		0.998	0.996	Yes
Empowerment	Seniors versus managers	1.000	0.999	Yes
Reputation		0.991	0.992	No
Leadership		1.000	1.000	Yes
Pay		0.997	0.995	Yes
Conditions		0.999	0.998	Yes
Satisfaction		1.000	1.000	Yes
Loyalty		0.993	0.998	No

PLS-MGA test. For the model associated with the hybrid segmentation variable levels, we compared path coefficients for each group using the PLS-MGA test. Pairwise group comparisons allowed us to investigate specific differences between groups of employees, obtaining the results shown in Table 10. Since the PLS-MGA is a one-tailed test according to Hair et al. (2018), the p-values indicated whether a path coefficient was significantly larger for one group than for another group, while the 1-p values indicated whether there was a significant difference in the opposite direction. We found that empowerment was significantly more important for seniors than for juniors or managers ($p=0.999$ and $p<0.001$, respectively), that company reputation was significantly more important for managers than for juniors or seniors ($p=1.00$ in both cases), that leadership was significantly more important for juniors than for seniors or managers ($p=0.001$ and $p<0.001$, respectively), that pay was significantly more important for juniors than for managers ($p=0.020$) and, finally, that work conditions were more important for managers than for juniors or seniors ($p=0.978$ and $p=0.996$, respectively).

Table 9 MICON step 3 results

Construct	Equality of composite values					
	Mean values			Variance values		
	Mean diff.	2.5–97.5%	Verified	Log. var ratio	2.5–97.5%	Verified
<i>Junior versus manager</i>						
Empowerment	0.198	[−0.080–0.078]	No	0.076	[−0.132–0.132]	Yes
Reputation	0.045	[−0.080–0.084]	Yes	−0.019	[−0.126–0.122]	Yes
Leadership	0.242	[−0.078–0.079]	No	−0.054	[−0.110–0.109]	Yes
Pay	−0.458	[−0.079–0.077]	No	0.153	[−0.124–0.113]	No
Conditions	0.381	[−0.079–0.080]	No	−0.138	[−0.116–0.112]	No
Satisfaction	−0.139	[−0.078–0.079]	No	0.548	[−0.157–0.154]	No
Loyalty	0.055	[−0.080–0.085]	Yes	0.054	[−0.144–0.136]	Yes
<i>Junior versus senior</i>						
Empowerment	0.563	[−0.100–0.100]	No	−0.153	[−0.146–0.148]	No
Reputation	0.587	[−0.100–0.100]	No	−0.055	[−0.133–0.134]	Yes
Leadership	0.320	[−0.098–0.100]	No	−0.190	[−0.136–0.139]	No
Pay	−0.225	[−0.099–0.101]	No	−0.088	[−0.133–0.133]	Yes
Conditions	0.603	[−0.098–0.100]	No	−0.216	[−0.144–0.142]	No
Satisfaction	0.494	[−0.098–0.098]	No	−0.037	[−0.148–0.152]	Yes
Loyalty	0.715	[−0.101–0.099]	No	−0.068	[−0.142–0.141]	Yes
<i>Senior versus manager</i>						
Empowerment	−0.411	[−0.078–0.080]	No	0.237	[−0.125–0.125]	No
Reputation	−0.543	[−0.082–0.081]	No	0.045	[−0.118–0.112]	Yes
Leadership	−0.086	[−0.079–0.080]	No	0.130	[−0.115–0.112]	No
Pay	−0.210	[−0.082–0.080]	No	0.213	[−0.123–0.116]	No
Conditions	−0.243	[−0.080–0.084]	No	0.080	[−0.114–0.106]	Yes
Satisfaction	−0.717	[−0.080–0.082]	No	0.573	[−0.149–0.141]	No
Loyalty	−0.681	[−0.079–0.081]	No	0.122	[−0.125–0.124]	Yes

Table 10 PLS-MGA results according to the hybrid segmentation variable

Path coefficient	Path			<i>p</i>		
	JNR	SNR	MNG	JNR versus SNR	JNR versus MNG	SNR versus MNG
Empowerment → Satisfaction	0.513	0.650	0.477	0.999*	0.173	0.000*
Reputation → Satisfaction	0.088	0.113	0.210	0.763	1.000*	1.000*
Leadership → Satisfaction	0.155	0.025	0.037	0.001*	0.000*	0.639
Pay → Satisfaction	0.143	0.101	0.072	0.140	0.020*	0.161
Conditions → Satisfaction	0.121	0.103	0.187	0.311	0.978	0.996*
Satisfaction → Loyalty	0.671	0.658	0.633	0.326	0.063	0.146

*Significant differences. *JNR* juniors, *SNR* seniors, *MNG* managers

5 Discussion

Our article describes a practical approach to the problem of dealing with heterogeneity in PLS-SEM when the number of segmentation variables is high, making it difficult to determine the most significant differences. In our approach we identify the most heterogeneous segments according to the segmentation variables available and then verify significant differences between the segmented groups.

While MGA is a well-established method for analysing heterogeneity in PLS-SEM models, its usefulness becomes limited when the number of comparisons starts to increase, making it difficult to suitably interpret results. This problem arises when we are dealing with either/both a large number of segmentation variables and segmentation variable levels. Pathmax analysis is a more suitable approach for a large number of variables, as the algorithm is capable of detecting the most significantly different groups; however, it has the drawback that there is no *a posteriori* significance test for the path coefficients of the estimated models. Our hybrid multigroup PLS-SEM merges the pathmax algorithm with classical MGA in such a way as to overcome the main limitations of both techniques. This hybrid approach also makes it easier, using the pathmax tree results, to define the hybrid segmentation variable and consider interactions between categorical variables.

Since both MGA and pathmax analysis are well established techniques, the method does not represent any substantial theoretical innovation, yet its contribution is eminently practical in terms of offering a way to test differences in path coefficients when dealing with a large number of categorical variables.

We demonstrated application of our hybrid approach using a practical example of job satisfaction and loyalty in a banking context, where it is important to design effective HR management practices according to different employee profiles. Applying our hybrid method to the six segmentation variables of gender, age, marital status, education, job level, and antiquity, we were able to define three specific employee groups, namely, junior employees, senior employees, and managers. For these groups, furthermore, we determined which of empowerment, company reputation, pay, work conditions, and leadership were especially important drivers of satisfaction and loyalty, finding that leadership and pay were significantly more important for juniors, empowerment for seniors, and company reputation and work conditions for managers. Furthermore, job level and antiquity were found to be the most important factors in segmenting employees. This kind of knowledge is valuable in that it can be used to shape more specific, and therefore, more effective, HR management policies.

This approach to segmentation has applications that go beyond HR management, as it could potentially be applied in many areas of business and marketing where identifying different groups and understanding why they are different is crucial to delimiting target groups and defining good strategies. It also has recently been applied in the psychology field (Signore et al. 2019; Macchitella et al. 2020).

As a practical example in relation to psychology, researchers could apply the hybrid multigroup PLS-SEM to the employability model proposed by Signore et al. (2019) and use it to test whether significant differences exist according to the hybrid segmentation variables based on the categorical variables of age, educational level, disciplinary area, and working conditions.

Regarding when the hybrid multigroup PLS-SEM approach can be used (see Table 11), researchers should consider aspects such as the application context, the source of heterogeneity, the complexity of the research, and the type of hypothesis to be tested. The hybrid

Table 11 When to use hybrid multigroup PLS-SEM

Criteria	Yes	No
Research context	Exploratory	Confirmatory
Heterogeneity source	Observed	Not observed
Analytical complexity	> = 2 segmentation variables, many levels	< 2 segmentation variables
Test hypothesis	Open research question	Specific theoretical hypothesis

multigroup PLS-SEM approach is suitable when the primary goal of the research is exploratory and heterogeneity is observed, i.e., when, in analysing different potential sources of heterogeneity, hypotheses cannot be defined a priori regarding differences between groups and when the hybrid segmentation variables are defined after the pathmox analysis. This approach is also indicated when the complexity of the analysis is high, as, due to either/both a large number of segmentation variables and segmentation variable levels, between-group comparisons and interpretation of differences are complex. The approach is not indicated for confirmatory research, unobserved heterogeneity and testing a specific hypothesis.

Another limitation concerns the way in which the hybrid segmentation variables are defined. The method uses pathmox partitioning criteria based on the classical parametric F-statistic, and so assumes the normality of perturbation terms with equal variance in all the endogenous constructs, yet this is an assumption rarely met in practice. Nonetheless, although violations of this assumption may lead to an invalid F-statistic, Lamberti et al. (2016, 2017) demonstrated the robustness of the test.

From a computational point of view the hybrid approach is more efficient than the classical multigroup approaches, because the multigroup test and MICOM procedure are applied only to the most significant partitions determined according to the pathmox analysis, which means that fewer comparisons are necessary. The importance of this aspect is directly related to the number of observations and the number of comparisons.

While an objective of this research was to provide practical guidelines on dealing with heterogeneity in PLS-SEM when the number of segmentation variables is high (and so overcome the limitations of pathmox and MGA analyses), a limitation of our research is that we did not perform a simulation study to confirm the potential of this hybrid procedure.

A final point to underline is that results will vary according to the depth of the pathmox tree. However, in a PLS-SEM context, any increase in tree depth will be directly correlated with increased results complexity and greater difficulty in their interpretation. While more complex solutions can be considered according to the interpretability of the results, we recommend a depth of two, corresponding to a maximum of four terminal nodes and six multigroup comparisons.

6 Conclusion

We have described a practical approach to dealing with heterogeneity when the number of categorical variables is high and the research context is exploratory. Our hybrid multigroup PLS-SEM overcomes several limitations associated with classical MGA (incapacity to deal with a large number of segmentation variables or levels and mixed effects of the segmentation variables) and with pathmox analysis (the lack of an a posteriori test of significant

differences in path coefficients). As well as providing guidance on how to apply the procedure, we describe an application to real data for an international bank. More specifically, we investigated the most significant drivers of satisfaction and loyalty in employees (empowerment, company reputation, pay, work conditions, and leadership) according to heterogeneity factors (age, gender, marital status, education, job level, and antiquity) that potentially affect perceptions of the work climate. The three distinct groups of employees identified, namely, juniors, seniors, and managers, were found to perceive the work climate in different ways: leadership and pay were more motivational for junior employees, empowerment for seniors, and company reputation and work conditions for managers. This information is potentially useful for HR managers in profiling employee groups and designing specific HR policies.

Appendix

Constructs/indicators	Mean	SD	Skewness	Kurtosis
<i>Empowerment</i>				
Emp1—Recognition of work performed	3.459	0.894	−0.615	0.067
Emp2—Employee treatment as responsible	3.515	0.832	−0.555	0.23
Emp3—Teamwork is empowered	3.291	0.92	−0.502	−0.172
Emp4—Autonomy is favoured	3.239	0.863	−0.348	−0.098
Emp5—Confidence in performed tasks	3.48	0.853	−0.435	0.168
Emp6—Creativity and initiative are endorsed	3.231	0.821	−0.373	0.31
<i>Reputation</i>				
Rep1—Organization's reputation	4.215	0.686	−0.51	0.135
Rep 2—Organization's values	4.165	0.743	−0.72	0.791
Rep 3—Organization's customer relationships	3.949	0.718	−0.613	1.386
Rep 4—Organization's internal relationships	3.914	0.716	−0.371	0.429
Rep 5—Organization's external projection	3.959	0.738	−0.54	0.91
<i>Pay</i>				
Pay1—Salary	3.823	0.748	−0.817	1.22
Pay2—Social benefits	3.672	0.768	−0.49	0.347
Pay3—My salary corresponds to my duties	3.12	0.915	−0.278	−0.171
Pay4—My salary corresponds to my effort	3.111	0.932	−0.326	−0.525
<i>Conditions</i>				
Cond1—Enough personnel in the office	3.242	1.016	−0.463	−0.597
Cond2—Enough time to perform the tasks	2.924	0.961	−0.151	−0.901
Cond3—Conditions and tools for work	3.693	0.769	−0.922	1.188
<i>Leadership</i>				
Sup1—Agenda and planning	3.489	1.042	−0.594	−0.253
Sup2—Receptiveness	3.914	1.055	−0.966	0.382
Sup3—Encouraging	3.607	1.054	−0.567	−0.248
Sup4—Communication	3.326	1.101	−0.376	−0.573
Sup5—Celebrating success	3.761	1.132	−0.803	−0.102
<i>Satisfaction</i>				
Sat1—Overall rating of satisfaction	4.037	0.699	−1.043	2.646

Constructs/indicators	Mean	SD	Skewness	Kurtosis
Sat2—Tasks in accordance with capabilities	3.537	0.855	−0.927	0.506
Sat3—Possibility to know efficiency	3.595	0.768	−0.762	0.696
Sat4—Possibility to learn new things	3.624	0.839	−0.754	0.512
Sat5—Usefulness of performed job	3.86	0.658	−0.867	2.157
Sat6—Fulfilment of expectations	3.489	0.858	−0.857	0.374
<i>Loyalty</i>				
Loy1—I am unwilling to leave in case of not finding alternative	3.586	0.908	−0.74	0.467
Loy2—I am committed to the institution	4.134	0.789	−1.03	1.796
Loy3—I trust in the proper direction of the management	3.84	0.825	−0.651	0.779

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Availability of data and material The data that support the findings of this study are not available due to legal policy restrictions.

Code availability The SmartPLS 3.2 software and the genpathmox package in R software version 0.4 provided the model estimates (global and multigroup analysis) and the pathmox analysis, respectively.

Compliance with ethical standards

Conflict of interest The author declares that he has no conflict of interest.

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